

Introduction

Large Language Models (LLMs) like BERT have significantly advanced natural language processing but come with substantial computational costs. BERT-base, with approximately 109 million parameters, is often too resource-intensive for edge deployment or real-time applications. The project focuses on the re-implementation of the paper “DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter” by Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf.

The primary contribution of the original paper is the application of knowledge distillation during the pre-training phase. By using a “teacher” model (BERT-base) to train a “student” model (DistilBERT), the authors produced a model with 40% fewer parameters that retained 97% of its teacher's performance while being 60% faster during inference.

Chosen Result

We wanted to recreate the key results from the paper which includes:

- 1) Parameter reduction: can we show that distillation significantly reduces parameter count like what is shown in Table 3?
- 2) Downstream performance: can we achieve 92.81% accuracy on the downstream IMDb task that was shown in Table 2?
- 3) Inference speedup: can we achieve the ~1.63x inference time speedup that was shown in Table 3?
- 4) Ablation studies: we did an ablation study where we removed the Cosine Embedding Loss. Will we see the same significant loss in accuracy that was shown in Table 4?

Reproducing these specific results is critical because they quantify the efficiency-performance trade-off that defines DistilBERT’s core value proposition

Methodology

Our re-implementation focused on reproducing the DistilBERT architecture and training pipeline with modifications suited to our available compute resources.

We implemented two student architectures: a **6-layer student** (standard DistilBERT) and a **3-layer student** (to further test even more aggressive compression). Following the paper’s methodology, the student models were initialized by taking every n th layer from the BERT-base teacher (every 2nd layer for the 6-layer model; every 4th for the 3-layer model).

The original paper utilized the full Toronto BookCorpus and English Wikipedia. Due to compute limitations, we used a 10% subset of the BookCorpus (approx. 740k examples) for pre-training. Data was tokenized using the standard BERT tokenizer with a maximum sequence length of 128 (truncated if data was too long, padded if it was too short).

We implemented a custom DistillationTrainer using the triple loss objective outlined in the paper.

- **Distillation Loss** ($\alpha_{ce} = 0.5$): Soft cross entropy loss between the student and the teacher logits with a temperature of $T=2$.
- **MLM Loss** ($\alpha_{mlm} = 0.2$): Standard masked language modeling loss.
- **Cosine Embedding Loss** ($\alpha_{cosine} = 0.3$): Encourages the student’s hidden states to align with the teacher’s

For downstream evaluation, we fine-tuned both student models and the teacher on the **IMDb Sentiment Classification** dataset for 3 epochs with a learning rate of 2×10^{-5} and a batch size of 16. We evaluated our reimplementation of DistilBERT through IMDb accuracy, inference speed, and parameter count evaluation metrics.

Results & Analysis

Model	Parameters	Reduction	IMDb accuracy	Latency
BERT-base teacher	109.48M	0.00%	93.46% (paper)	7.55 / 7.34 ms
6-layer student, full loss	66.96M	38.84%	91.42%	4.24 ms
6-layer student, CE ablated	66.96M	38.84%	91.40%	same architecture; not separately benchmarked
3-layer student, full loss	45.69M	58.27%	90.01%	2.38 ms
3-layer student, CE ablated	45.69M	58.27%	90.12%	same architecture; not separately benchmarked

Our 6-layer model achieved a **40% parameter reduction**, matching the original paper exactly. We observed a **1.82x speedup**, which exceeds the 1.6x (60% faster) reported in the paper; this difference is likely due to our use of shorter sequence lengths (128 vs 512) and different hardware (NVIDIA T4/MPS).

The absolute IMDb accuracy of our 6-layer student (90.1%) was lower than the 97% teacher-retention rate cited by Sanh et al. This discrepancy is likely attributed to the reduced pre-training data (10% of BookCorpus vs. full BookCorpus + Wikipedia) and the limited number of pre-training epochs (1 epoch).

Additionally, our ablation study failed to confirm the paper’s findings that removing the Cosine Embedding loss resulted in a significant drop in accuracy. As we can see in the table, the CE ablated models perform either marginally worse than the models which used the full loss function, or even in one case better (the 3-layer CE ablated model performed better than the full loss model). We believe that the reason for this result is that our downstream task was a very simple binary sentiment classification task (compared to the original paper which used more complicated inferences tasks like the GLUE tasks).

The 3-layer model demonstrated that even aggressive compression (60% parameter reduction) maintains surprising utility, achieving 87.6% accuracy while being over 3x faster than the teacher.

Reflections

The project provided several key insights into the mechanics of model compression. First, the importance of **weight initialization** cannot be overstated; starting from teacher layers allowed the model to converge significantly faster than training from scratch. Second, the **triple loss function** acts as a powerful regularizer, helping the student model retain the “shades of meaning” captured in the teacher’s soft probabilities.

A major challenge was the resource cost of pre-training. Even a 10% subset of one dataset required several hours of GPU time, highlighting why the original authors emphasized the “cheaper and lighter” nature of the final model for downstream users.

Future work should involve scaling the pre-training to the full Wikipedia corpus to see if the accuracy gap closes. Additionally, exploring **layer-selection strategies** beyond simple linear mapping (e.g., using the first k layers or searching for optimal layer combinations) could yield better performance for the 3-layer variant. Finally, evaluating the models on the GLUE benchmark would provide a more robust understanding of how compression affects linguistic generalization across diverse tasks.

References

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